

Sarbjot Singh

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EDUCATION

Acadia University — Wolfville, Nova Scotia

2024 – 2027

Bachelor of Applied Computer Science, Data Analytics, Co-op | GPA: 3.66 / 4.33

- Relevant coursework: Data Analytics, Statistics, Database Management Systems, Data Structures & Algorithms, Software Engineering

EXPERIENCE

Research Assistant – Sepsis Prediction System

March 2026 – Present

Acadia University · Wolfville, NS

- Processed 1.55M+ hourly ICU observations from 40,332 patients through a Python ML pipeline, expanding 40 raw clinical variables into 280 model features using temporal statistics, missingness indicators, forward filling, and train-only median imputation.
- Trained and evaluated 3 classification models — Logistic Regression, Random Forest, and XGBoost — on a patient-separated 70/15/15 split, achieving 0.830 AUROC and 0.112 average precision with XGBoost on 233,500 held-out observations from 6,050 patients.
- Improved XGBoost test average precision by 44.7% over Logistic Regression and packaged the workflow into a Streamlit dashboard and CLI inference tool for comparing 3 models, visualizing hourly risk scores, applying validation-selected thresholds, and exporting patient-level predictions.

PROJECTS

CanadaPulse — Canadian Economic & Labour Market Data Platform | [GitHub](#)

2026

Python · PostgreSQL · dbt · FastAPI · Next.js · Docker · GitHub Actions · Airflow

- Engineered an end-to-end batch ETL/ELT pipeline across 2 official data sources using Python, PostgreSQL, schema validation, deterministic keys, and data-lineage tracking to ingest and process Canadian labour-market and interest-rate data.
- Designed a dimensional data warehouse with 16 dbt models and 67 automated tests, achieving 0 test errors across staging, intermediate, dimension, fact, and analytics-mart layers while validating grain, referential integrity, and source reconciliation.
- Automated pipeline orchestration and reliability with a 4-task Apache Airflow DAG, GitHub Actions CI/CD, and 26 passing Python tests, implementing task dependencies, retries, scheduled refresh workflows, execution metadata, and data-quality monitoring.
- Built a PySpark batch-processing pipeline that transformed 13,860 monthly observations into 1,188 annual groups and generated partitioned Parquet outputs for scalable historical analytics.

ValleyCabs — Full-Stack Booking & Delivery Application | [GitHub](#)

2025

TypeScript · Python · FastAPI · PostgreSQL/PostGIS · Redis · WebSockets · Google Maps APIs · Stripe · Firebase

- Architected a TypeScript/Python full-stack platform for local transportation and delivery operations using FastAPI, PostgreSQL/PostGIS, Redis, and REST APIs, supporting 2 service workflows, 3 user roles, and 6 backend domains.
- Built a real-time GPS and navigation workflow tailored for short-distance local trips, using WebSockets, PostGIS, and Google Maps APIs to refresh vehicle coordinates every 3–5 seconds and expose live tracking 30 minutes before scheduled departure.
- Designed a local booking and payment workflow covering 4 transaction stages — authorization, confirmation, cancellation, and refund — using Stripe Payment Intents and webhooks, achieving booking/payment synchronization within 2–3 seconds.
- Engineered trip and delivery updates for local operations across 4+ service states, using Redis, WebSockets, and Firebase Cloud Messaging while achieving <500 ms API response times and 1–3 second notification delivery after backend status changes.

CERTIFICATIONS

Google Data Analytics Professional Certificate

Google / Coursera · 2025

SQL · Tableau · Data Cleaning · Data Validation · Data Visualization · Spreadsheets

Applied Software Engineering Fundamentals Specialization

IBM / Coursera · 2025

Python · Git/GitHub · Linux/Bash · Flask · REST APIs · Unit Testing · SDLC

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, TypeScript, Go

Frameworks: React, Next.js, FastAPI, Node.js, Flask, Streamlit

Data Engineering: ETL/ELT, Data Pipelines, dbt, Apache Airflow, PySpark, Data Warehousing, Dimensional Modeling, Batch Processing

Databases & Backend: PostgreSQL, PostGIS, MongoDB, MySQL, Redis, REST APIs, WebSockets

Data & ML: Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Tableau, Power BI

Tools & DevOps: Git, GitHub, Docker, GitHub Actions, CI/CD, Linux, Jupyter Notebook, Google Cloud Platform